

ANURAG KALAPALA

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Entry-level Machine Learning Engineer with hands-on experience developing ML models achieving 15-30% accuracy improvements across healthcare, finance, and text classification domains. Proficient in TensorFlow, PyTorch, and cloud deployment with expertise in computer vision, NLP, and statistical modelling. Strong foundation in end-to-end ML pipeline development from data preprocessing to production deployment. Seeking full-time opportunities to leverage AI/ML expertise in solving complex business challenges.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, C++, JavaScript

Machine Learning: TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Pandas, NumPy

Deep Learning: CNN, RNN, LSTM, Transfer Learning, ResNet, BERT, Transformers, GANs

NLP: NLTK, SpaCy, Hugging Face, Sentiment Analysis, Topic Modelling

Computer Vision: OpenCV, Image Processing, Object Detection, Medical Imaging Cloud Platforms: AWS, Azure, GCP

Tools: Git, Docker, Flask, Tableau, Power BI, Jupyter, MLflow

Databases: SQL, PostgreSQL, MongoDB, Time-Series Databases

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Welkin Technologies LLC | July 2024 - April 2025

- Developed time-series analysis models for wearable ECG data using Python and signal processing, improving arrhythmia detection accuracy by 15%
- Built fraud detection POC for auto insurance using ensemble ML algorithms, presenting findings to C-level executives
- Implemented real-time health monitoring pipeline processing 10,000+ wearable devices daily
- Created predictive models for patient health risk assessment enabling proactive healthcare interventions

Graduate Research Assistant | University of Massachusetts Lowell | January 2024 - May 2024

- Developed deep learning models for sentiment analysis using TensorFlow and PyTorch, achieving 20% accuracy improvement
- Fine-tuned BERT and RoBERTa models for biomedical NER, reducing error rates by 30%
- Published research findings at IEEE International Conference on Machine Learning Applications
- Collaborated on interdisciplinary NLP applications in healthcare and social media analytics.

TECHNICAL PROJECTS

Real-Time Fake News Detection Platform | Python, BERT, NLP, Hugging Face

- Built end-to-end detection system processing 17,500 PolitiFact articles across six truthfulness levels
- Fine-tuned BERT and SBERT models for multi-class misinformation classification
- Developed real-time monitoring dashboard with customizable alerts
- Implemented scalable pipeline processing 1,000+ articles per minute

Skin Lesion Classification System | PyTorch, ResNet-18, Medical Imaging

- Developed skin cancer classification using HAM10000 dataset with 10,015 dermoscopic images
- Fine-tuned ResNet-18 for seven-class classification achieving 83% test accuracy
- Implemented data augmentation and regularization techniques
- Created model interpretability features using Grad-CAM for clinical decision support

EDUCATION

Master of Science in Computer Science | University of Massachusetts Lowell | January 2023 - May 2024

Coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Data Mining

Bachelor of Engineering in Electronics and Communication | Chaitanya Bharathi Institute of Technology

Coursework: Signal Processing, Digital Communications, Data Structures, Statistics

CERTIFICATIONS

Deep Learning Specialization - DeepLearning.AI (2023)

TensorFlow Developer Certificate - Google (2023)

Machine Learning Engineering for Production - DeepLearning.AI (2023)